The Neurophysiology of Cybersickness: A Review of BCI-based Characterization and Prediction

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***Abstract*—Cybersickness remains a barrier to widespread virtual reality (VR) adoption. Brain-Computer Interfaces (BCI) and neuroimaging techniques offer promising means to objectively measure, predict, and mitigate this condition. This systematic review synthesizes current knowledge at the BCI-cybersickness intersection, identifying methodological challenges and proposing future research directions. A systematic literature search across PubMed, Web of Science, and Scopus identified 83 peer-reviewed studies following inclusion criteria. Studies were categorized and compared based on objectives, neuroimaging modalities, computational methods, and findings. Our synthesis highlights common neurophysiological signatures of cybersickness, with increased low-frequency (delta, theta) and decreased alpha-band power in EEG, and engagement of a cortico-subcortical network including the insula, cingulate cortex, and visual-vestibular regions in fMRI and fNIRS. Computational models, particularly deep learning approaches, demonstrate promise in classifying cybersickness states. Key research directions include predicting individual susceptibility from resting-state brain activity and developing closed-loop BCI systems that actively mitigate symptoms by modulating the VR environment in real-time. While the field has transitioned from correlational studies to predictive models, progress is hampered by methodological inconsistencies, inconsistent labeling, and limited generalizability. This review establishes a comprehensive overview of the literature essential for developing robust BCI applications for cybersickness in user-aware adaptive systems.**

***Index Terms*—Human-centered computing, Human computer interaction (HCI), Interaction paradigms, Virtual reality.**

I. Introduction

Virtual Reality (VR) technology has permeated numerous sectors beyond entertainment, becoming integral tools in fields such as healthcare, professional training, and education [1]. Generally, Virtual Reality (VR) is defined as a computer-generated simulation that enables users to perceive and interact with a three-dimensional environment through real-time multisensory feedback. A typical VR system includes a head-mounted display for stereoscopic rendering, motion-tracked controllers or sensors for user input, and a computing unit for environment simulation and rendering. However, there is no unifying definition of VR [2].

The successful application of VR thus hinges on its ability to deliver high-fidelity, convincing experiences. Despite this potential, a significant barrier to widespread adoption is the onset of cybersickness (CS) [3]. Cybersickness is a specific form of visually induced motion sickness (VIMS) [4] characterized by symptoms such as nausea, dizziness, and disorientation that arise during or after immersion in a virtual environment [3]. However, the same set of symptoms is also observed in related conditions such as motion sickness (MS) and simulator sickness (SS). While distinctions have been proposed, such as CS being more strongly associated with compelling self-motion (a phenomenon known as vection) and disorientation, in contrast to the oculomotor or nausea-dominant profiles of SS and MS [5], those distinctions are not always taken into account in the literature. Since these terms are often used interchangeably, this review considers findings across studies referring to Cybersickness, Visually Induced Motion Sickness, Simulator Sickness, and Motion Sickness, where relevant.

CS affects a substantial portion of the user population, with some studies reporting that 60% to 95% of participants experience some level of CS [3]. The prevalence of CS severely degrades the user experience, impairs task performance, and can lead to premature session termination [6]. The assessment of CS relies heavily on subjective, post-hoc measures like the Simulator Sickness Questionnaire (SSQ) [7]. However, such questionnaires are retrospective, prone to recall bias, and may not capture the dynamic nature of the sickness experience [8], [9]. This reliance on subjective reports creates a critical need for objective, real-time metrics to accurately track the user's state.

Brain-Computer Interfaces (BCIs) present a promising methodology for addressing this challenge. A BCI is a system that measures central nervous system activity and converts it into an artificial output that can replace, restore, enhance, supplement, or improve natural central nervous system output [10]. Within BCI research, a distinction is made between active, reactive, and passive systems [10]. Of particular relevance to CS research are passive BCIs, which utilize brain activity as an implicit source of information about the user's cognitive or affective state without the need for voluntary control. By continuously monitoring neurophysiological signals such as the electroencephalogram (EEG), passive BCIs can provide an objective, real-time window into the user's internal experience. This capability enables the development of a new class of user-aware adaptive systems that can respond dynamically to mitigate discomfort [11], [12].

This paper provides a systematic literature review and critical synthesis of the research at the intersection of BCI and cybersickness. The primary aims of this review are threefold. First, we synthesize the current state of the art, consolidating findings related to the neural correlates, prediction, and mitigation of CS using BCI methods. Second, we critically appraise the methodologies, experimental designs, and analytical techniques prevalent in the existing literature, highlighting significant challenges to generalizability and

replicability. Then, we identify research gaps, and finally highlight potential future effective BCI-driven applications for managing cybersickness.

II. Theoretical and Methodological Background

A. The Etiology of Cybersickness

The precise etiology of cybersickness remains a subject of investigation, with several theories proposed to explain its multifaceted nature. These theories are not always mutually exclusive and often provide complementary perspectives on the phenomenon.

The most dominant framework is the *Sensory Conflict Theory*, which posits that cybersickness arises from a discrepancy between information received from different sensory channels, particularly the visual, vestibular, and proprioceptive systems [13], [14]. Several specific hypotheses branch from this core theory. The *Vection conflict hypothesis* is more specific and states that the visual system needs to perceive vection while the vestibular and proprioceptive systems report a stationary or incongruent physical state to experience cybersickness [15]. The *Subjective Vertical Mismatch Theory* suggests a conflict between the perceived gravitational vertical and the one predicted by the brain based on visual cues [16]. The *Rest Frame Conflict Hypothesis* attributes sickness to ambiguity in identifying a stable, stationary frame of reference within the environment [17]. Furthermore, the *Poison Theory* offers an evolutionary explanation, proposing that the brain misinterprets the sensory mismatch as a sign of neurotoxin ingestion, triggering a defensive response that includes nausea and vomiting [18].

A second prominent framework is the *Postural Instability Theory*. This theory proposes that cybersickness is not a direct result of sensory mismatch, but rather a consequence of a prolonged inability to maintain stable postural control [19]. According to this view, the unfamiliar sensory dynamics of a virtual environment challenge the user's postural control system; the resulting instability, which can be measured objectively, precedes and predicts the subjective experience of sickness.

Other theories focus on more specific physiological mechanisms. The *Vergence-Accommodation Conflict* is a factor unique to stereoscopic displays like HMDs. It describes the mismatch between the distance at which the eyes converge to fuse a stereoscopic image and the fixed focal distance of the device's screen, leading to eyestrain and discomfort [20]. The *Eye Movement Theory* suggests that sickness is caused not by the sensory conflict itself, but by the specific patterns of eye movements (e.g., nystagmus, torsion) that are generated by the vestibular system in an attempt to compensate for the perceived motion [21].

More recently, these concepts have been integrated within a *Predictive Coding Framework* [22]. This view re-contextualizes sensory conflict as a failure of predictive processing. The brain is conceptualized as a prediction engine, constantly generating models of the world to anticipate sensory inputs. When there is a persistent and unresolvable prediction error, a large discrepancy between expected and actual sensory signals, the brain fails to update its internal model. This failure to reconcile the conflict is hypothesized to manifest as the aversive experience of cybersickness.

B. Neuroimaging Modalities for Cybersickness Research

While cybersickness assessment has traditionally relied on subjective measures such as questionnaires, the need for objective, real-time metrics has driven researchers toward neuroimaging approaches to measure corresponding brain activity. Electroencephalography (EEG) is the most widely adopted modality in the reviewed literature. Its principal advantages are its excellent temporal resolution, allowing for the tracking of rapid changes in neural dynamics, its portability, which facilitates studies in mobile and interactive VR paradigms [19], [23], as well as its relative affordability. Functional Near-Infrared Spectroscopy (fNIRS) is an emerging optical imaging technique that measures hemodynamic changes in cortical blood flow. It offers a practical balance between temporal and spatial resolution and is notably more robust to motion artifacts than Functional Magnetic Resonance Imaging (fMRI), making it well-suited for cybersickness research involving movement [12], [24]. Finally, fMRI provides unparalleled spatial resolution, enabling the precise localization of activity in both cortical and subcortical structures. However, it also has a lower temporal resolution and requires the subject to be immobile, which can be problematic for the study of cybersickness. Despite these limitations, fMRI has been instrumental in identifying the core brain networks underlying nausea and sensory integration, such as the insular and cingulate cortices [25], [26].

III. Review Methodology

A. Search Strategy

A systematic literature search was conducted on March 27, 2025, to identify all relevant peer-reviewed articles. The search was performed across three major academic databases: PubMed, Web of Science, and Scopus. No date restrictions were applied to ensure a comprehensive historical overview of the field. The following search query, designed to capture studies at the intersection of BCI and cybersickness, was utilized:

```
(Brain Computer Interface OR EEG OR
    electroencephalogram OR neuromarkers OR
    MEG OR fMRI OR ECoG OR fNIRS)
AND
(Cybersickness OR ((virtual reality OR VR) AND
     motion sickness) OR simulator sickness OR
     visually induced motion sickness OR
    Virtual Reality Motion Sickness)
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B. Study Selection and Eligibility Criteria

The initial search yielded a total of 360 articles. Following the removal of 119 duplicate entries, 241 unique records remained for screening. This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020

statement, an evidence-based guideline for transparent and complete reporting. This framework enables readers to appraise the review's methodological quality and to ensure the reproducibility of the search and selection strategy. The full selection process is detailed in the PRISMA flow diagram (Figure 1).

Each of the 241 records underwent a manual screening process based on titles and abstracts to assess relevance according to predefined eligibility criteria. The inclusion criteria for this review were as follows:

- The study employed a neuroimaging or BCI methodology (e.g., EEG, fMRI, fNIRS) to measure brain activity.
- The study focused on cybersickness or a closely related term (e.g., VIMS, simulator sickness) within a VR or simulated environment.
- The study presented original empirical data from a human-subject study.

Conversely, articles were excluded if they met any of the following criteria:

- The work was non-empirical, such as literature reviews, meta-analyses, editorials, letters, or conference abstracts without a corresponding full paper.
- The full text of the article was unobtainable.

This screening process resulted in the exclusion of 155 articles, leaving 91 for full-text assessment. An additional two relevant articles were identified through manual citation searching and included. Of these 93 articles, 10 were subsequently removed due to the unavailability of their full text. This resulted in a final corpus of 83 peer-reviewed articles for detailed analysis.

C. Data Extraction and Thematic Synthesis

For each of the 83 included articles, a structured data extraction process was performed to systematically capture and catalogue key information. This process was guided by a predefined protocol designed to ensure consistency and facilitate subsequent comparison and synthesis. The following data points were extracted from each paper:

1) **Summary of Contribution:** A concise summary of the study's primary objective and its main contribution to the field of BCI-based cybersickness research.
2) **Cybersickness Measurement:** The methods used to measure or label cybersickness, distinguishing between subjective questionnaires (e.g., SSQ, FMS) and objective physiological or behavioral metrics (e.g., galvanic skin response, postural sway).
3) **BCI Methodology:** A detailed breakdown of the neuroimaging pipeline, including: (a) the measurement modality (e.g., EEG, fMRI), (b) the key feature extraction techniques (e.g., power spectral density, event-related potentials, functional connectivity), and (c) the primary analysis or classification methods (e.g., statistical tests, machine learning models).
4) **Theoretical Framework:** The explicitly stated or implicitly assumed theory of cybersickness (e.g., Sensory Conflict, Postural Instability) that guided the study's design or interpretation of results.
5) **Quantitative Performance:** Reported performance measures, such as classification accuracy, correlation coefficients (e.g., PLCC, SROCC), root mean square error (RMSE), or statistical significance values.
6) **Empirical Study Details:** For empirical studies, specific details of the experimental protocol, including the VR environment, display hardware, cybersickness-inducing stimulus, and the number of human subjects.
7) **Reported Shortcomings:** Any limitations, caveats, or weaknesses explicitly identified by the authors or apparent from the study's design.

Following data extraction, a quantitative analysis was conducted to structure the review and identify overarching trends, methods, and results. To facilitate this synthesis, each paper was assigned to a primary research objective category. An initial set of objective categories was defined *a priori* based on established research paradigms in the field. This set was iteratively refined during the data extraction process to accurately capture the full spectrum of research goals present in the literature. The final categories, which form the organizational structure of the subsequent results section, include: Classification (Binary, Likert, Continuous), Correlates (e.g., Brain Regions, Rhythms, Physiological), Mitigation (A/B Testing, Real-time), User Differences, Dataset Provision, Recovery, and studies on the Effects of Cybersickness on other cognitive or performance metrics.

IV. Results

The selected studies address several distinct objectives, which we categorize as follows. These include: (1) Observed study characteristics, (2) the identification of neurophysiological and physiological correlates of the cybersickness state; (3) the development of computational models for the classification and prediction of sickness; (4) the characterization of neurophysiological factors underlying individual susceptibility; (5) the testing of BCI-driven mitigation and intervention strategies; and (6) ancillary investigations encompassing recovery dynamics, dataset contributions, cognitive performance effects, and hardware-specific neural response patterns. The research demonstrates temporal evolution in research focus, with early investigations concentrating primarily on correlates and classification of cybersickness, while recent research has diversified into additional fields of study, as illustrated in Figure 2.

The following section synthesizes the findings from this literature, structured thematically according to these primary research objectives.

A. Study Characteristics

The synthesis encompasses 83 empirical studies spanning from 2001 to 2025, with a notable acceleration in research activity observed in recent years: 59.0% of studies were conducted from 2021 onwards. The total participant population across all investigations comprises 2213 individuals, with studies employing a mean of 26.7 ± 25.7 participants per investigation (median: 20, range: 6-202). Age demographics show a mean of 24.0 ± 2.3 years (median: 23.7, range:

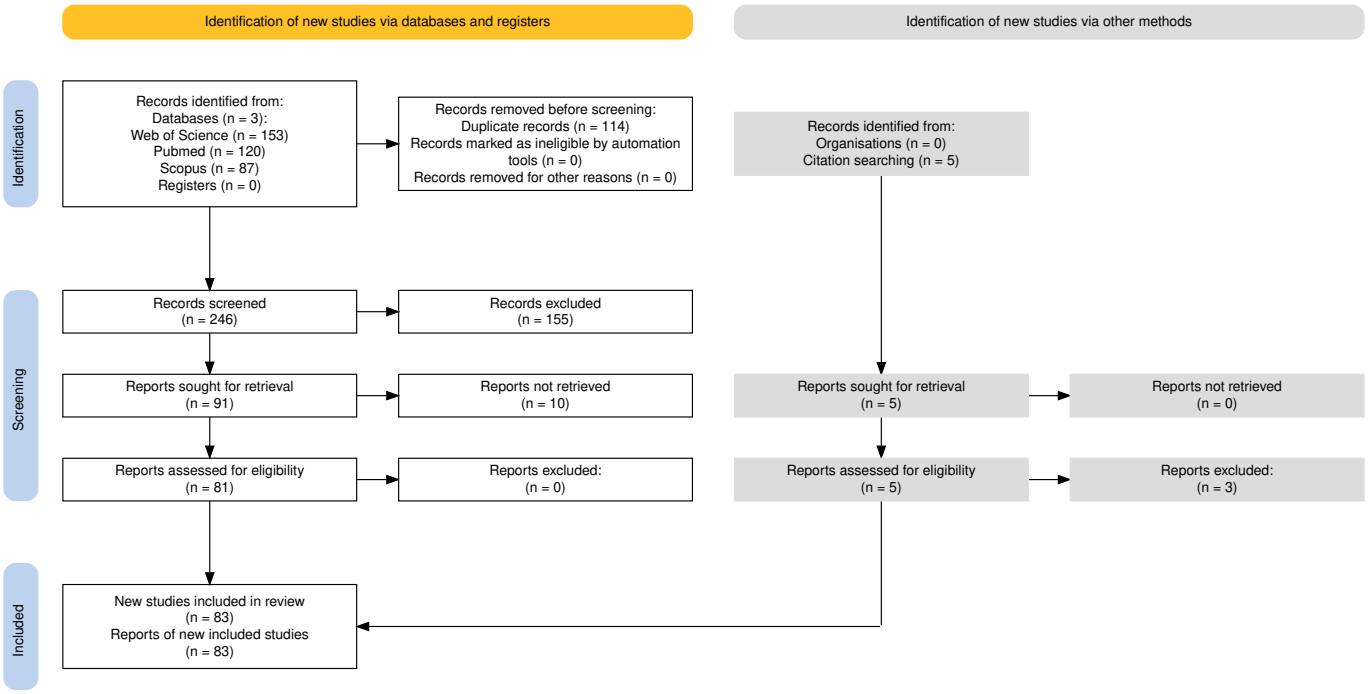


Fig. 1: PRISMA flow diagram illustrating the study identification, screening, eligibility, and inclusion process.

20.4-29.9) across the 49 studies that reported participant age, reflecting a predominant focus on young adult populations. An overview of age distributions across all studies is presented in Figure 5.

1) Neuroimaging and Physiological Recording Modalities: EEG represents the predominant neuroimaging modality, employed in 89.2% of studies (74 of 83), followed by fNIRS at 6.0% (5 studies) and fMRI at 4.8% (4 studies). EEG configurations demonstrate substantial variability, with a mean of 38.9 ± 56.4 channels (median: 30, range: 1-256). The distribution reveals a predominance of standard montages, with the most frequently employed configurations including 32-channel (21 studies), 14-channel (7 studies), 64-channel (7 studies), 256-channel (4 studies), and 30-channel (4 studies) setups. A visual representation of the distribution of EEG channel configurations across studies is presented in Figure 3.

Beyond primary neuroimaging modalities, studies frequently incorporate additional physiological measurements. Multi-modal approaches are employed in 55.4% of investigations, with electrocardiography (ECG) being the most common supplementary modality (15.1% of studies), followed by joystick input (12.8%) and head tracking (11.6%). Other recorded signals include galvanic skin response (GSR) (8.1%), electrooculography (EOG) (8.1%), eye tracking (5.8%), photoplethysmography (PPG) (4.7%), skin temperature (4.7%), electrogastrography (EGG) (3.5%), button box (3.5%), heart rate monitor (3.5%), and respiration monitor (3.5%). Additional modalities encompass motion capture systems (2.3%), blood pressure monitor (2.3%), balance board (2.3%), and various other devices including Kinect, pulse oximeter, electromyography (EMG), keyboard, force plate, mouse, and accelerometer (each 1.2%).

2) Experimental Design and Data Collection Protocols: Study designs predominantly employ within-subject comparisons (75.9%), with 7.2% employing between-subject designs exclusively, and 16.9% utilizing both within-subject and between-subject approaches. Session protocols demonstrate considerable variation, with the majority implementing single-session designs and the remainder conducting multiple sessions (mean: 1.4 ± 0.8 sessions, range: 1-4). Experimental sessions last a mean of 49.8 ± 34.9 minutes (median: 48.5, range: 2-180), while cybersickness-inducing stimulus exposure durations average 17.2 ± 17.8 minutes (median: 10.0, range: 0-90).

Virtual reality exposure is delivered through head-mounted displays in 56.0% of studies, screen-based displays in 15.5%, projection systems in 19.0%, other configurations in 6.0%, and television displays in 3.6%. Virtual environments encompass navigation tasks (24.4%), driving simulations (20.0%), video content (15.6%), roller coaster experiences (13.3%), games (7.8%), and other scenarios (16.7%). The relationship between VR device types and virtual environments is illustrated in Figure 6.

3) Cybersickness Assessment Methods: Cybersickness measurement employs questionnaire-based approaches in 67.0% of studies, continuous rating methods in 17.6%, Likert scale assessments in 8.8%, and other methodologies in 6.6%. The Simulator Sickness Questionnaire (SSQ) represents the predominant assessment instrument, utilized in 63.4% of studies, followed by the Fast Motion Sickness Scale (FMS) (9.8%) and Motion Sickness Susceptibility Questionnaire (MSSQ) (4.9%). Continuous input mechanisms include joystick-based ratings (48.0% of continuous studies), other methods (44.0%), and keyboard input (8.0%).

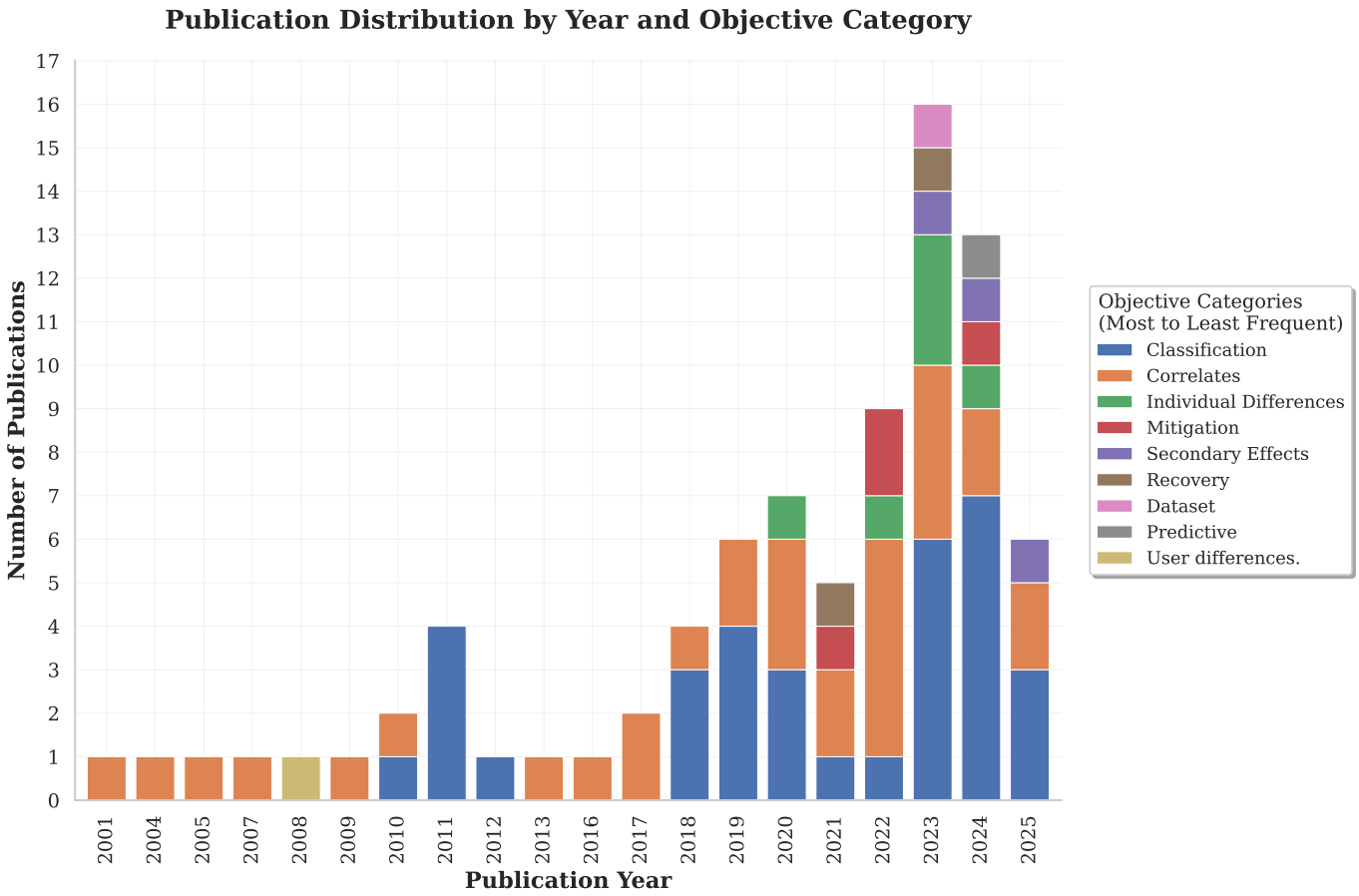


Fig. 2: Distribution of reviewed papers by year and research objective category

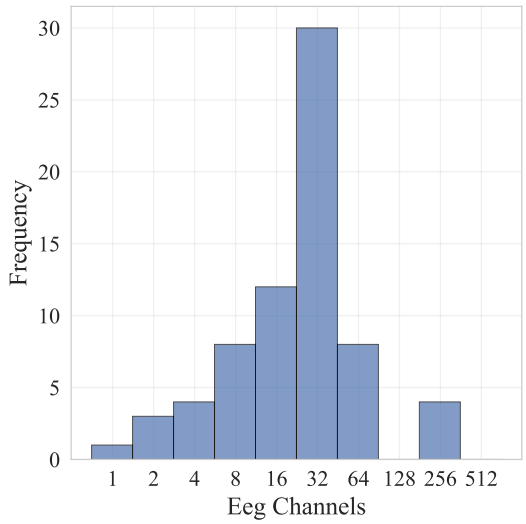


Fig. 3: Distribution of EEG channel configurations across cybersickness studies

4) Data Processing and Signal Analysis: EEG signal preprocessing demonstrates considerable heterogeneity across studies in terms of filtering approaches and frequency ranges. Band-pass filtering is employed across a wide spectrum of frequency ranges, from very low frequencies (0.01 Hz) to high gamma ranges (100 Hz), with the majority of studies applying filters in the 1-50 Hz range. The distribution of band-pass filter configurations across studies is presented in Figure 4, illustrating the substantial variability in preprocessing approaches employed in the literature.

B. Neural and Physiological Correlates of Cybersickness

A primary objective across a substantial portion of the literature has been to identify objective, physiological markers that co-vary with the subjective experience of cybersickness. This research addresses the question of what physiological changes occur when a user becomes sick. The studies have identified a set of neural and physiological correlates, which form the basis for the development of predictive models.

1) EEG Spectral Power and Connectivity: The most frequently reported neural signature of cybersickness is a modulation of EEG spectral power. Across numerous studies employing diverse VR stimuli and hardware, a pattern emerges characterized by an increase in low-frequency power and a corresponding decrease in alpha-band activity. Quantitative analysis across studies examining specific brain rhythms reveals patterns accross the literature. Delta rhythm changes show increases in 79.3% of studies, decreases in 10.3%, mixed findings in 6.9%, and no change in 3.4%. Theta

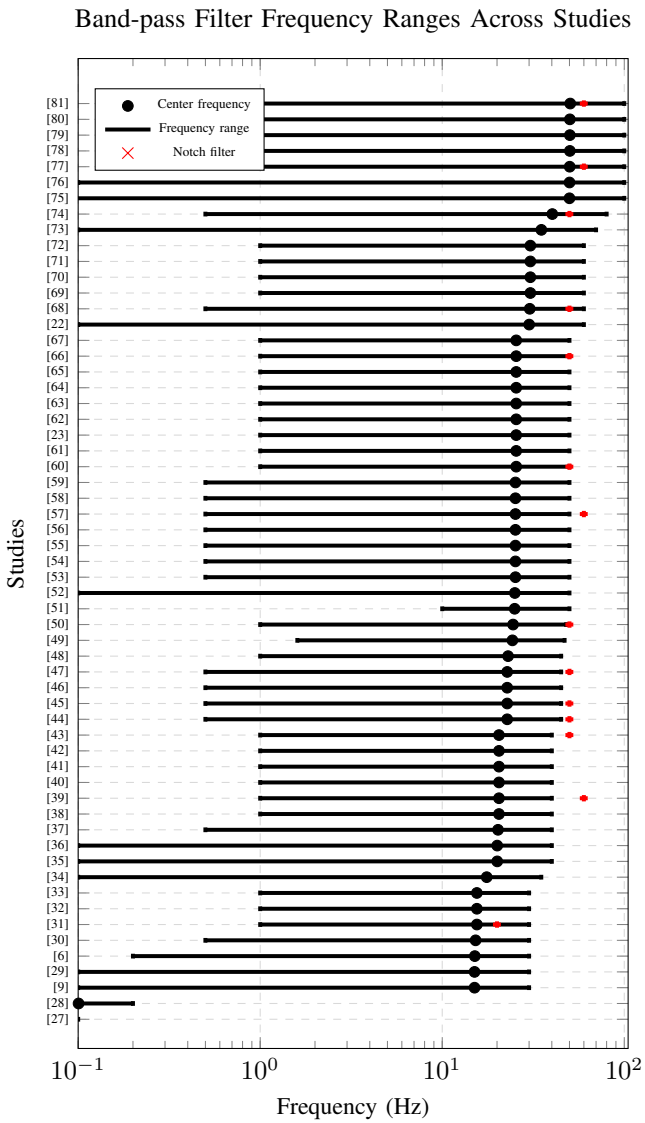


Fig. 4: Band-pass filter frequency ranges across studies showing center frequency (black dots) with frequency ranges (black error bars) and notch filter frequencies (red crosses). The frequency range shows the low and high cutoff frequencies of the band-pass filters, while red crosses indicate notch filter frequencies used to remove specific interference (e.g., 50/60 Hz power line noise).

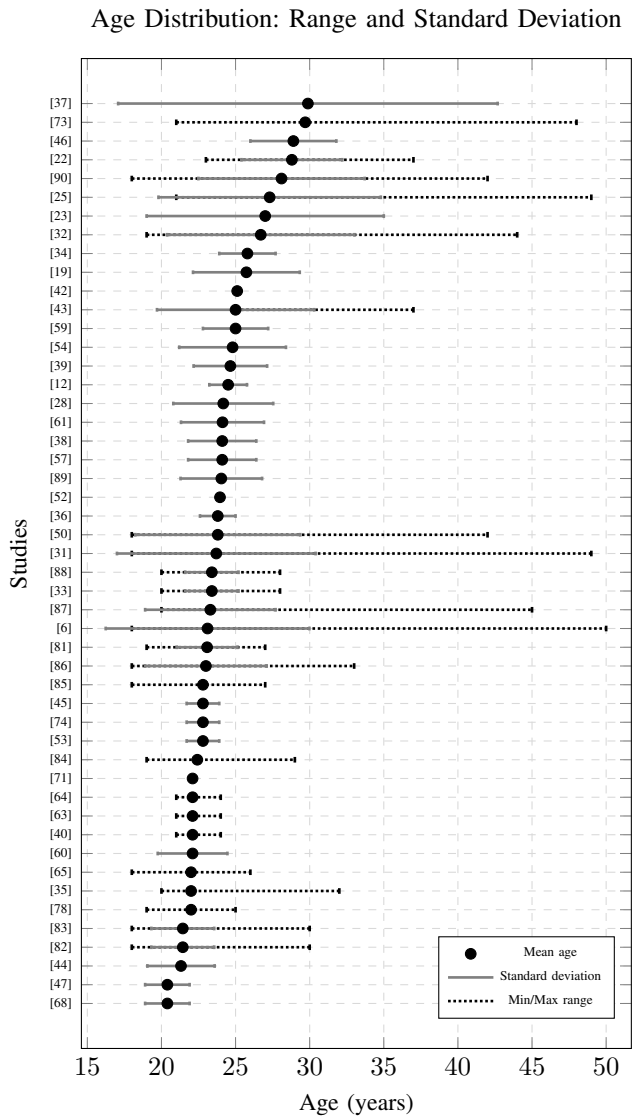


Fig. 5: Age distribution across studies showing mean age (black dots) with age ranges (dotted black error bars) and standard deviations when available (solid gray error bars). The age range shows the minimum and maximum participant ages, while the standard deviation shows the typical spread around the mean age.

rhythm investigations report increases in 73.3% of studies, with decreases in 13.3%, mixed results in 10.0%, and no change in 3.3%. Specifically, an increase in power within the delta (1-4 Hz) and theta (4-8 Hz) frequency bands has been associated with the onset and increasing severity of CS [22], [23], [32], [52], [81]. Concurrently, a commonly observed phenomenon is a suppression of power in the alpha band (8-13 Hz), particularly over central, parietal, and occipital cortices [19], [51], [63]. Alpha rhythm findings demonstrate greater variability, with 45.7% reporting increases, 31.4% showing decreases, 14.3% presenting mixed results, and 8.6% finding no change. This alpha suppression is often interpreted as a marker of increased cognitive load, potentially reflecting the brain's attempt to manage sensory conflict or maintain postural control [19]. While less consistent, changes have also been reported in higher frequency bands, with some studies noting decreases in beta power [81] and others reporting increases [79] or its utility for channel selection [42]. Beta rhythm studies show increases in 41.4% of investigations, no change in 24.1%, decreases in 24.1%, and mixed findings in 10.3%. Gamma rhythm research reports increases in 40.0% of studies, no change in 33.3%, decreases in 20.0%, and mixed results in 6.7%. A comprehensive analysis of rhythm changes across different brain regions is presented in Figure 7. The test-retest reliability of these spectral markers has been formally assessed, with one study demonstrating good reliability for delta, theta, and alpha power changes in frontal and central

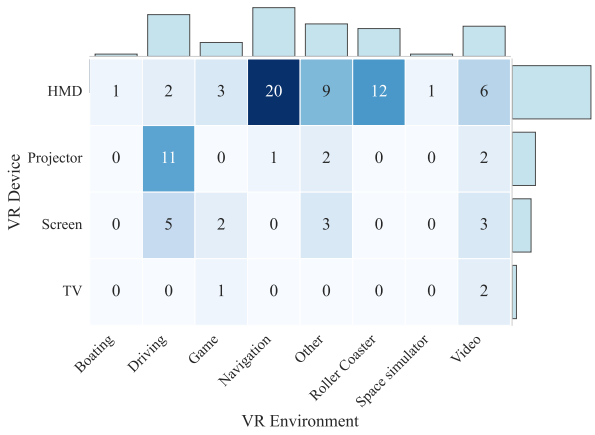


Fig. 6: Heatmap of the distribution of VR device types and virtual environments across studies. Studies employing multiple devices or environments contribute towards each.

areas across multiple sessions [59].

Furthermore, the normalization of delta wave power has been shown to serve as an objective marker for the dissipation of symptoms during the recovery phase [57].

In addition to band power, researchers have explored more complex features derived from the EEG signal. These include metrics such as spectral entropy and skew [53], Gravity Frequency, and Kolmogorov Complexity [91], [92], as well as amplitude modulation features, which one study identified as significant for CS characterization [93]. Others have used the mean frequency of the theta band as a direct indicator of motion sickness level [40]. Investigations into functional connectivity provide information on more systemic CS-related brain dynamics. Functional connectivity investigations reveal mixed findings in 56.2% of studies, increases in 18.8%, decreases in 18.8%, and no change in 6.2%. One finding is a general decrease in information transfer, measured via transfer entropy, with the onset of VIMS [22]. More specifically, studies using EEG microstate analysis have shown that CS alters the temporal dynamics of canonical brain networks and is associated with decreased functional connectivity between the visual cortex (V5) and vestibular processing regions [38]. Connectivity-based features are also used as direct inputs for deep learning models, often based on metrics like the Phase-Locked Value (PLV) [44], [80]. The causal relevance of these spectral patterns is suggested by intervention studies where symptom mitigation correlates with a normalization of these same EEG bands [37], [61].

2) Hemodynamic and Regional Brain Activity (fNIRS & fMRI): EEG offers high temporal resolution, whereas fMRI and fNIRS provide high spatial localization, identifying the specific brain regions involved in the cybersickness experience. fNIRS and fMRI each represent 4.8% of studies, reflecting their complementary but less prevalent role compared to EEG-based investigations. fMRI studies have delineated a ”nausea network”. One study dissociated the temporal evolution of nausea, identifying a network including the amygdala and putamen that was active preceding a transition to

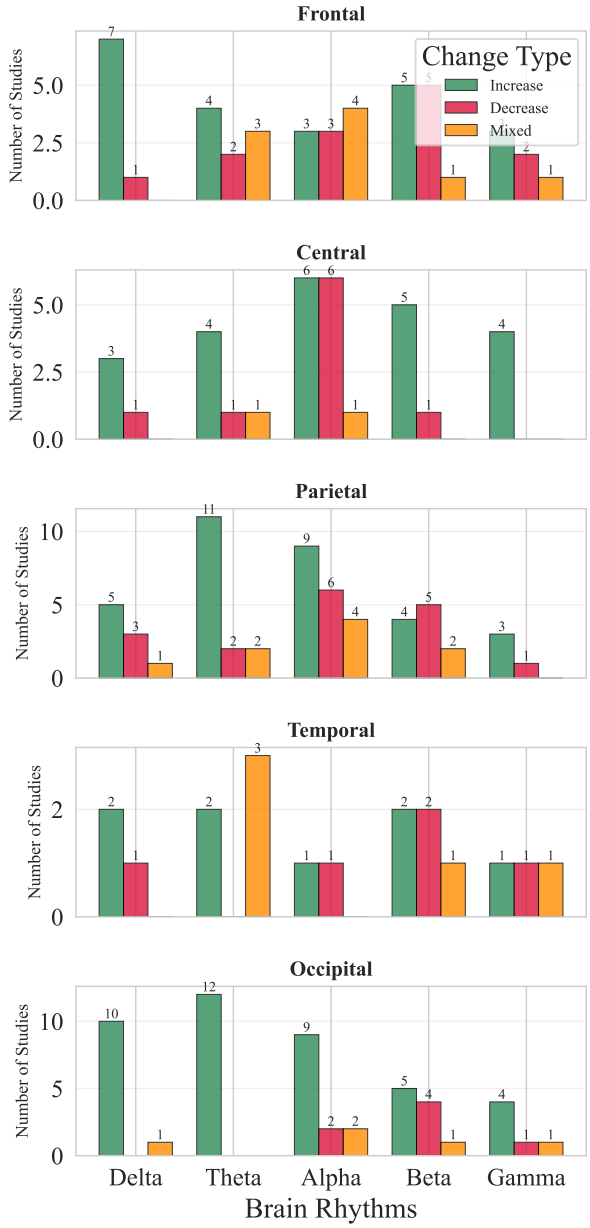


Fig. 7: Brain rhythm changes across anatomical regions during cybersickness

stronger nausea, and a broader network including the insular, anterior cingulate, and orbitofrontal cortices that was active following the transition [25]. Furthermore, motion sickness has been shown to increase the functional connectivity between the visual motion processing area (MT+/V5) and nausea-associated regions like the insula and mid-cingulate cortex [26]. The involvement of this network is highlighted by findings that its functional connectivity is also modulated during the post-exposure recovery phase [27]. This central processing has been linked to peripheral physiology, with activity in distinct brain networks correlating with sympathetic and parasympathetic autonomic outflow during nausea [94].

fNIRS studies, which offer greater portability, have corroborated and extended these findings. Research using

fNIRS has identified deactivation in the bilateral angular gyrus [28] and activation changes in the occipital lobe [24] and parieto-temporal regions [95] that correlate with subjective sickness scores. Others have linked an increase in total hemoglobin concentration directly to cybersickness [12], [96]. This body of work indicates a complex hemodynamic response, with evidence for both increased regional perfusion in some cortical areas and a decrease in global cerebral perfusion during cybersickness episodes [95].

3) Event-Related Potential (ERP) and Evoked Potential: Beyond ongoing oscillations, a subset of studies has used ERPs to probe how cybersickness impacts discrete cognitive processes. These studies indicate that cybersickness is accompanied by measurable changes in cognitive function. One finding is that VR sickness is associated with a reduction in the amplitude of the P3b component in a dual-task paradigm, a classic marker of diminished attentional resources [6]. Relatedly, VIMS has been shown to impair response inhibition, as reflected by a larger N2 amplitude and a smaller, delayed P3 component in an oddball task [34]. Other work has shown that trait-like susceptibility to cybersickness is associated with distinct baseline ERP characteristics, such as excessive N2 amplitudes [49] and higher P3 amplitudes during sensory mismatch conditions [73]. An approach using heartbeat-evoked potentials (HEPs) has linked the subjective experience of disorientation to a significant suppression of fronto-central HEP amplitudes, providing a potential link between interoceptive processing and a key symptom of cybersickness [39].

4) Other Physiological Correlates: Many studies have adopted a multimodal approach, recording other physiological signals alongside neuroimaging data. This work indicates that cybersickness is a whole-body phenomenon. Electrogastrography (EGG) has been identified as a measure for detecting nausea-specific symptoms [87]. Measures of autonomic nervous system arousal, such as heart rate variability (HRV) and galvanic skin response (GSR), are also consistently modulated by cybersickness and have been integrated into predictive models [58], [94], [97]. In a study that integrated VR with a motion platform, electromyography (EMG) signals were reported to be more informative for classifying motion sickness than the concurrently recorded EEG signals [36]. Finally, electrooculography (EOG) has been used to show that an increased eye blink rate correlates with subjective cybersickness reports [46].

C. Classification and Prediction of Cybersickness

A substantial body of research has focused on using the previously identified neural correlates to automatically classify or predict a user's cybersickness state. This line of inquiry seeks to determine if these physiological changes can be reliably detected and quantified by computational models. The research objective is to develop objective, real-time assessment systems that can operate continuously and implicitly during VR use. Analysis of computational approaches across the literature reveals that deep learning architectures are employed in 15 studies (34.1%), traditional machine learning classifiers in 14 (31.8%), regression models in 13 (29.5%), thresholding in 1 (2.3%), and unsupervised approaches in 1 (2.3%). We defined statistical and correlational analysis as studies that identify and validate statistically significant relationships between physiological signals and cybersickness without building predictive models, including methods such as t-tests, ANOVA, and correlation analyses. Traditional machine learning was defined as studies employing established algorithms with engineered features for discrete state classification, including Support Vector Machines, Linear Discriminant Analysis, and k-Nearest Neighbors. Deep learning encompasses studies utilizing multi-layered neural networks for automatic feature learning from raw data, including Convolutional Neural Networks, Recurrent Neural Networks, and hybrid architectures. Regression models were defined as studies predicting continuous cybersickness severity values using methods such as Support Vector Regression and linear regression approaches. The distribution of these classification approaches is illustrated in Figure 8, with detailed breakdowns of specific methods within each category shown in Figure 9. Additionally, 55.4% of studies adopt multi-modal approaches, combining EEG with other physiological measurements. The most common additional modalities are Electrocardiography (ECG) and Head tracking, which are used in 15.1% and 11.6% of studies, respectively. Model validation strategies predominantly employ k-fold cross-validation (65.7%), with additional approaches including train-test splits (14.3%), other methods (11.4%), and leave-one-subject-out cross-validation (8.6%). Performance evaluation reveals mean classification accuracies of 92.5% ± 8.6% for within-subject validation (range: 73.3%-99.9%) and 86.8% ± 9.8% for cross-subject validation (range: 68.0%-100.0%).

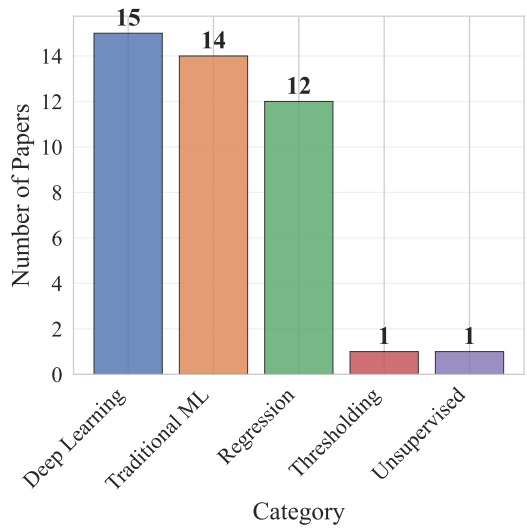


Fig. 8: Distribution of classification method categories across cybersickness studies

1) Methodological Approaches and Feature Engineering: The methodologies for automated cybersickness detection have evolved in parallel with advances in machine learning. One set of approaches relies on traditional machine learning

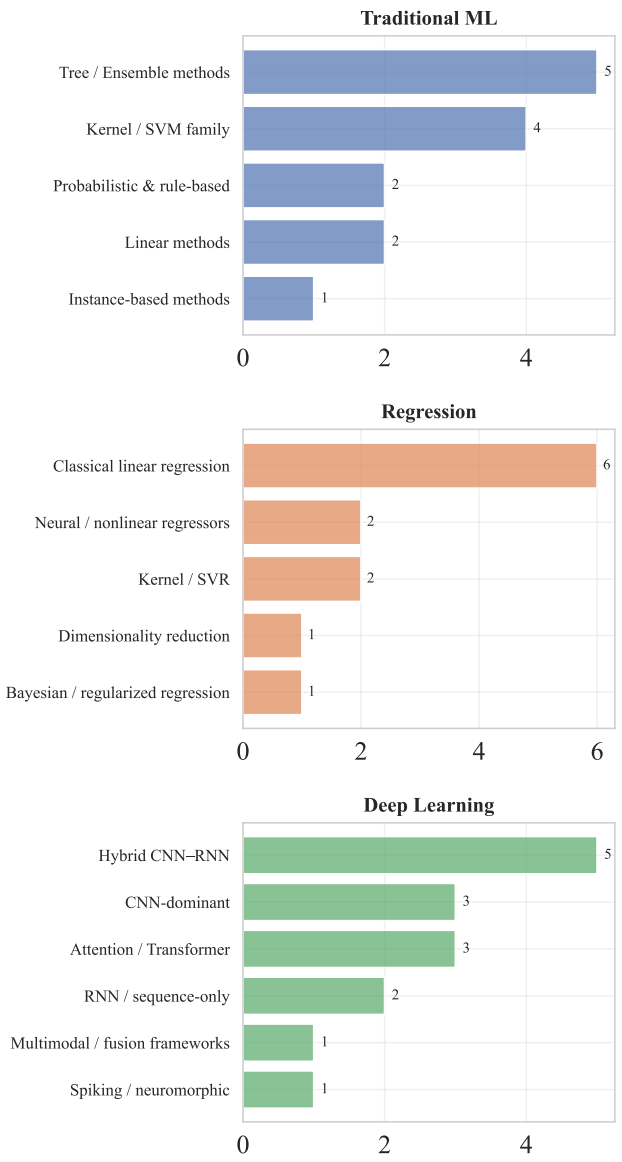


Fig. 9: Detailed breakdown of specific methods within each classification category

classifiers fed with pre-selected, or ”handcrafted,” features. These models include Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Linear Discriminant Analysis (LDA) [47], [67]. Features extracted from EEG for these models are frequently based on Power Spectral Density (PSD), wavelet-derived energy ratios [30], or spectral entropy and skew [53]. Other work has employed time-domain statistical features such as signal variance [98]. To reduce dimensionality and improve model performance, some studies focus on feature selection algorithms, using methods like Correlation Feature Selection (CFS) [42] and genetic algorithms [72].

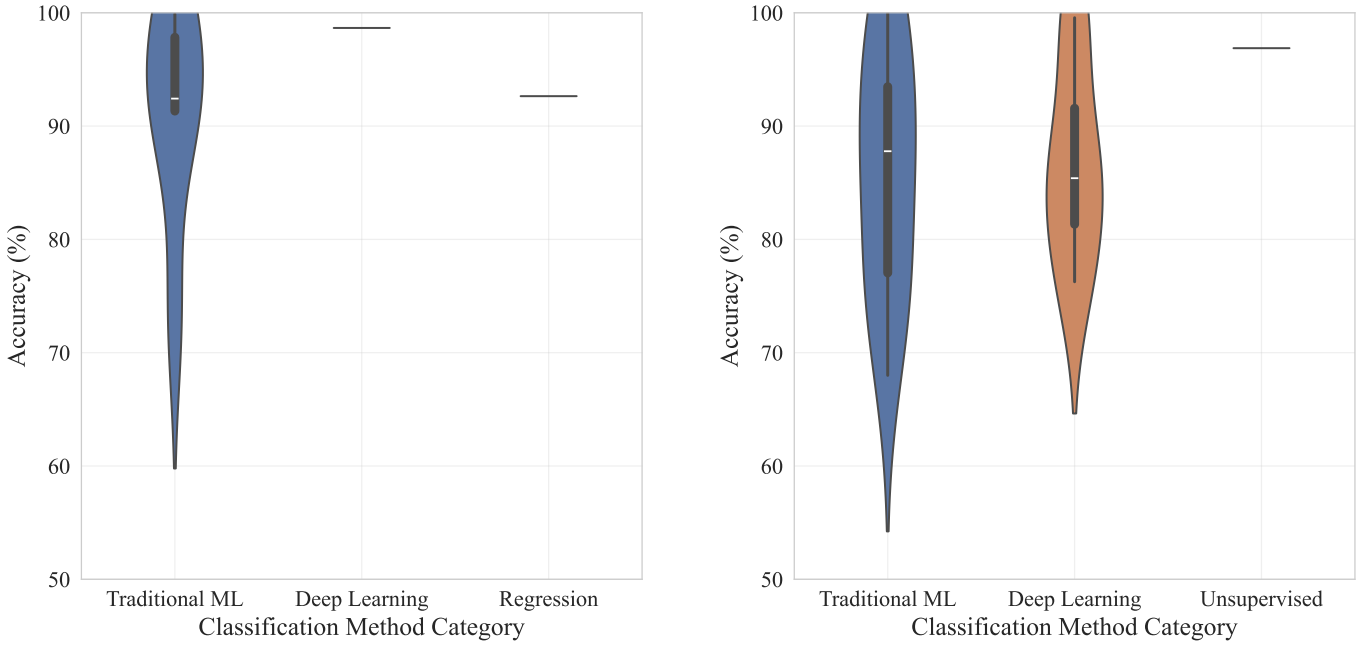
A second set of approaches utilizes deep learning architectures, which can learn relevant features directly from the data. Convolutional Neural Networks (CNNs) are widely used to extract spatial and spectral features from EEG time series, spectrograms, or topological scalp maps [48]. To capture the temporal evolution of cybersickness, these are often combined with Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM) [99] or Gated Recurrent Units (GRU) [55]. Other explored architectures include spiking neural networks [86] and specialized shallow CNNs designed for real-time detection in mitigation systems [11]. More recent models frequently employ hybrid architectures, such as combining CNNs and LSTMs [44] or implementing dual-pathway CNNs that process raw EEG and functional connectivity data in parallel [80]. These models may be enhanced with attention mechanisms to weigh the importance of different channels or time points [78]. The inputs to these models have also diversified, incorporating functional connectivity matrices [44] or features such as entropy and cross-frequency coupling asymmetry [68].

Beyond EEG-only models, a number of studies have demonstrated the utility of multimodal data fusion. This involves combining neural signals with peripheral physiological data, such as galvanic skin response (GSR) and blood volume pulse (BVP) [97], or with non-neural signals like electromyography (EMG) from postural muscles, which was found to be highly informative [36]. Other modalities, including fNIRS, have also been used as input for regression models [96]. A different approach involves knowledge transfer, where a model trained on EEG data is used to enable prediction based solely on features extracted from the VR video content [77]. To improve model interpretability, some studies have incorporated explainable AI (XAI) techniques [97].

2) Classification Tasks and Performance: Model performance is typically evaluated across three main tasks. The most common task is binary classification, which distinguishes between a ”sick” and ”not sick” state. In within-subject validation paradigms, many studies report accuracies exceeding 95% [44], [48], [67], [68], [78], [83]. Quantitative analysis of reported performance metrics reveals that within-subject models achieve a mean accuracy of 92.5% ± 8.6% (median: 92.7%, range: 73.3%-99.9%). Cross-subject models, which are trained on a group of users and tested on a new user, yield accuracies typically in the range of 68.0%-100.0% [30], [55], with a mean accuracy of 86.8% ± 9.8% (median: 86.7%). The accuracy distributions for both validation approaches are compared in Figure 10.

A second task is multi-level classification, which categorizes cybersickness into several discrete levels of severity. Studies have demonstrated three-level classification using multimodal data fusion [9], [85] and EEG alone, for instance with a rule-based induction algorithm [41] or a genetic algorithm with an SVM [62].

A third task is continuous prediction or regression, where the objective is to predict a continuous value, such as a total SSQ score. Performance is measured by correlation coefficients and error metrics. Deep learning models have demonstrated high correlations between predicted and actual SSQ scores [45], [54], [56], [66]. Earlier work in this area used methods such as Support Vector Regression (SVR) [69], [70] and linear regression [71]. Current architectures include Siamese LSTMs [74] and frameworks that fuse physiological



(a) Within-subject accuracy

(b) Cross-subject accuracy

Fig. 10: Classification accuracy distributions by method category for within-subject and cross-subject validation approaches

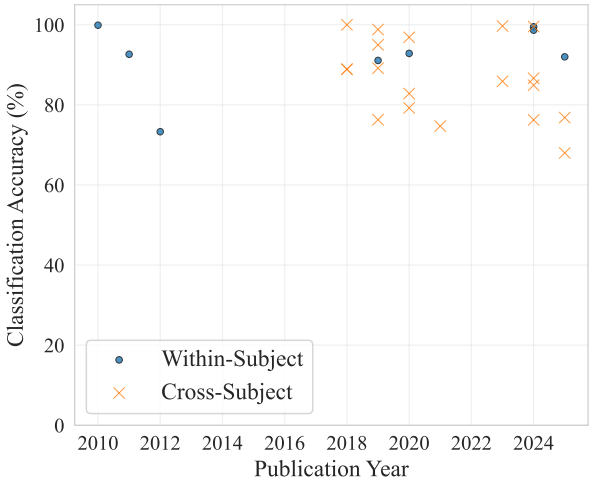


Fig. 11: Classification accuracy over time comparing within-subject and cross-subject validation approaches

signals with VR video features [58]. Addressing the challenge of relying on discrete, post-hoc subjective labels, some research has focused on developing methods for continuous, calibration-free detection directly from physiological data [43]. The temporal evolution of classification performance across both validation approaches is depicted in Figure 11.

D. Individual Differences and Susceptibility

A well-established observation in cybersickness research is the significant inter-subject variability in symptom onset and severity. This has motivated a specific line of inquiry focused on identifying the neurophysiological basis of individual susceptibility. The goal of this research is to understand why some individuals are highly vulnerable to cybersickness while others remain resistant, and to determine if these differences can be predicted from brain activity.

A key finding in this area is that an individual's susceptibility to cybersickness can be predicted from their resting-state brain activity, even before any VR exposure. One study, using a spiking neural network, demonstrated the ability to predict whether a user would later become sick with 85.9% accuracy, based solely on pre-exposure resting-state EEG [86]. Other work has identified specific features of the resting-state signal as being predictive. For instance, one study found that EEG beta power in the left parietal region was a significant predictor of subsequent motion sickness ratings in an in-car VR platform [29]. Another EEG study found that aperiodic components of the signal predicted an individual's susceptibility to rotational motion [87]. These findings suggest the existence of pre-existing neural traits that predispose individuals to cybersickness.

Beyond prediction, studies have also identified distinct neural signatures that differentiate susceptible and resistant individuals during a nauseogenic experience. In a comparison between groups, resistant individuals were found to exhibit enhanced theta activity in the left parietal cortex, a potential biomarker of resilience [35]. Conversely, susceptible individuals show unique patterns of brain activity. One study found that VIMS-susceptible participants exhibited excessive parietal N2 ERP amplitudes and impaired theta-band phase synchronization compared to their resistant counterparts, suggesting a systematic impairment in dynamic cortical coordination [49]. Similarly, another ERP study found that individuals with high sickness levels presented with higher

P3 amplitudes during sensory mismatch conditions, indicating a greater demand on cognitive resources [73]. Analyses of spectral power and source-space activity have further revealed differential patterns of activation between susceptible and non-susceptible groups during vection [60]. For instance, one study noted that while the cybersickness condition increased delta and decreased alpha power in both sensitive and non-sensitive groups, the sensitive group showed significantly lower absolute power in the beta and gamma bands at baseline [52]. An early study also found that the ratio of theta to total EEG power in the frontal and parietal lobes differed significantly between sick and non-sick groups during a long-duration driving task, suggesting it as a potential physiological index of sensitivity [33].

In addition to neurophysiological traits, demographic factors have been shown to contribute to susceptibility. Specifically, several studies have consistently reported a higher incidence and severity of cybersickness in female participants compared to male participants [75], [76]. Together, this research demonstrates that individual susceptibility to cybersickness is not random but is rooted in measurable differences in baseline brain function, neural responses to sensory conflict, and demographic variables.

E. Mitigation and Intervention Strategies

A primary applied objective of cybersickness research is the development of mitigation and prevention strategies. A number of studies are focused on this area, developing and testing systems that use neurophysiological signals to trigger interventions. This research can be divided into two categories: real-time adaptive systems that modify the VR environment, and interventions that aim to directly modulate the user's brain state.

1) Real-Time Adaptive Systems: One approach involves creating closed-loop systems that can detect the onset of cybersickness and dynamically adapt the virtual environment to reduce its nauseogenic properties. These systems differ from static VR experiences by creating environments that are responsive to the user's internal state. Only two studies in the corpus implemented this approach, indicating that this area of study remains in early-stage development. One study developed a system that uses a two-stage shallow convolutional network to first detect cybersickness from EEG and then to infer a likely causal factor [11]. The system then selectively modifies the identified parameter (e.g., navigation speed or scene complexity) in real-time, resulting in a significant reduction in user-reported sickness compared to control conditions. Another approach utilized fNIRS to monitor total hemoglobin concentration in the brain [12]. Upon detecting an increase in hemoglobin correlated with sickness, the system dynamically reduced the user's field of view (FOV), demonstrating the feasibility of an fNIRS-based closed-loop system for symptom alleviation.

2) Neurostimulation, Neurofeedback, and Other Interventions: A second category of mitigation research investigates the efficacy of external interventions while measuring their effect on brain activity. Five studies in our corpus have taken this approach. One study demonstrated that transcutaneous auricular vagus nerve stimulation (taVNS) significantly reduced motion sickness symptoms [31]. The reduction in symptoms was correlated with taVNS-induced neural activation in specific brain regions, as measured by EEG source localization, providing a neurophysiological correlate for the treatment's effect. Another study explored Vibro-motor Reprocessing Therapy (VRT) and found its application was associated with significant reductions in alpha, delta, and theta spectral power [37]. Beyond stimulation, modifying other sensory inputs has also been shown to be effective. One study found that providing synchronized auditory and physical motion cues during a simulated bicycle ride reduced VIMS scores compared to conditions with mismatched or absent cues, an effect that was reflected in concurrent EEG recordings [61].

Some protocols leveraged neurofeedback (NF), where users are trained to volitionally modulate their own brain activity. However, research in this area has identified a significant confounding variable: the presence of cybersickness symptoms can directly impair a user's ability to successfully perform the neurofeedback training task [76]. Further complicating the study of such interventions, another experiment found that a placebo transcranial direct current stimulation (tDCS) procedure, while having no significant effect on reported sickness symptoms, paradoxically degraded performance on a concurrent NF task [75]. These findings illustrate a critical challenge for BCI-based interventions, where the physiological state being targeted can interfere with the user's ability to engage with the intervention itself.

F. Additional Findings: Recovery, Datasets, Cognition, and Confounding Factors

Beyond the primary research thrusts of correlation, prediction, and mitigation, several studies address additional aspects relevant to cybersickness. These include: (1) the neurodynamics of recovery after exposure, (2) publicly released datasets that support benchmarking and reproducibility, (3) the effects of cybersickness on cognition and task performance, and (4) identification of confounding factors such as hardware platform and virtual environment type that systematically influence neural response patterns.

1) The Neurodynamics of Recovery: While the majority of research focuses on the onset and progression of cybersickness, a subset of studies has investigated the neural processes that underlie recovery after the nauseogenic stimulus is removed. One study sought to objectively quantify the duration of recovery using both subjective SSQ reports and objective EEG measures [57]. The authors found that both measures indicated a recovery period of at least 11.5 minutes, with EEG delta wave power serving as a reliable objective marker of the recovery time course. Investigating the underlying network dynamics with fMRI, another study found that the recovery period from VIMS was characterized by enhanced resting-state functional connectivity patterns involving the insular, cingulate, and visual cortical regions, which are the same networks implicated in the emergence of

sickness [27]. This body of literature suggests that recovery is not a passive return to a baseline state but an active neurophysiological process.

2) Public Datasets: The advancement of robust and generalizable machine learning models is dependent on the availability of large, high-quality, and publicly accessible datasets. Several research groups have contributed to the field by providing such resources. One dataset was released by [77], containing EEG data from over 200 subjects experiencing 44 different VR contents, which was used to develop a model for predicting sickness from video features. Another contribution is a dataset containing EEG and subjective measures collected while participants used ten different VR locomotion techniques, providing a resource for studying the impact of interaction design on cybersickness [82]. More recently, a dataset was released alongside a method for continuous, calibration-free cybersickness detection, addressing the need for data with dynamic, real-time ground-truth labels rather than post-hoc questionnaires [43]. These open datasets are necessary for benchmarking new algorithms and fostering reproducible science.

3) Effects of Cybersickness on Cognition and Performance: A final ancillary theme explores the consequences of cybersickness on a user's ability to perform tasks within the virtual environment. This research demonstrates that cybersickness is an issue of both comfort and functional impairment. Studies using event-related potentials (ERPs) have provided direct neurophysiological evidence of this impairment. For instance, one study found that VR sickness severity was associated with a reduction in the P3b component amplitude, a classic ERP marker of diminished attentional resources available for a secondary task [6]. Relatedly, VIMS has been shown to impair executive functions such as response inhibition, an effect indexed by a larger N2 amplitude and a smaller, delayed P3 component during an oddball task [34].

This functional impairment also presents a specific challenge for BCI applications. In a pair of studies investigating VR-based neurofeedback (NF), it was found that the presence of cybersickness symptoms significantly hampered the user's ability to successfully perform the NF training task [76]. These studies also found that a placebo tDCS treatment, while having little effect on sickness symptoms, paradoxically impaired NF success, and consistently observed that female participants experienced more cybersickness and had worse NF performance than their male counterparts [75]. These findings highlight a challenge for BCI applications in VR: the physiological state being targeted can interfere with the user's ability to engage with the intervention itself.

4) Confounding Factors: Hardware Platform and Virtual Environment Effects: Meta-analysis of neural response patterns reveals differences based on hardware platform and experimental conditions. Head-mounted display studies demonstrate distinct alpha band response patterns compared to non-HMD configurations, with HMD studies showing alpha increases in 20.9% of cases while non-HMD studies show alpha decreases in 29.2% of cases, suggesting different underlying neural mechanisms. Mann-Whitney U analysis with Cliff's delta effect size revealed a statistically significant difference in alpha rhythm changes between HMD-based and screen-based cybersickness studies (p=0.044, Cliff's δ=0.455, large effect). This large effect size indicates that VR display technology substantially modulates alpha frequency responses during cybersickness, providing empirical support for device-specific neural response patterns that justify separate analytical treatment of HMD versus screen-based VR paradigms. Stimulus duration analysis reveals that HMD studies employ significantly shorter exposure periods (mean: 11.6 ± 16.6 minutes) compared to non-HMD studies (mean: 23.7 ± 15.0 minutes; U=242.0, $p < 0.001$, Cohen's $d = 0.76$), potentially reflecting comfort limitations with immersive displays or accelerated symptom onset in HMD environments. Virtual environment type demonstrates additional specificity in neural responses, with driving simulations exhibiting increased alpha responses 46.7% of the time while navigation tasks exhibit increased alpha only 15.8% of the time, potentially reflecting the distinct sensorimotor demands of driving scenarios.

V. Discussion

This work follows a recent scoping review by [8], which provides an overview of the use of EEG in cybersickness research. Their work, based on a literature search conducted in September 2021, systematically analyzed 33 EEG-based studies. The review reported that frequency or time-frequency analysis was the predominant method for feature extraction. Key neurophysiological findings identified in their analysis included an increase in delta power and a decrease in beta power associated with cybersickness, while results for the alpha band were noted as being inconsistent across studies. The authors also reported that classification models achieved accuracies between 79% and 100%, and that studies frequently used head-mounted displays with portable EEG systems and recruited participants primarily in their 20s. The volume of research in this domain has increased substantially since the conclusion of their search. Our analysis includes 49 additional studies published after their search date.

A. Methodological Challenges and Critical Perspective

While the reviewed literature demonstrates significant progress in applying BCI methods to cybersickness research, a critical appraisal reveals substantial methodological challenges. These issues temper the interpretation of many findings and present considerable barriers to the generalization, replication, and practical application of the reported results. This section will highlight four main challenges identified in the literature (1) optimistic accuracy estimates, (2) the lack of standardized cybersickness thresholds, (3) protocol disparities and (4) conceptual variability.

A prominent trend in the classification literature is an "accuracy arms race," with a proliferation of studies reporting exceptionally high performance. Numerous papers claim binary classification accuracies exceeding 95%, and in some cases, approaching 99% [44], [68], [78], [80], [83]. However, these impressive figures are often a product of experimental designs that limit their generalizability.

Many are achieved using within-subject validation on small, homogeneous participant pools, which may not translate to new, unseen users. Furthermore, the choice of data used for classification also plays a role. For example, some high-performing models compare resting-state EEG data collected before and after VR exposure [47], [78], a paradigm that, while useful for identifying state changes, is ill-suited for the real-time detection required by adaptive systems.

The foundation of any supervised learning model is the quality of its "ground truth" labels, and in cybersickness research, this foundation is ill-defined. The vast majority of predictive models are trained on subjective data, most commonly post-exposure SSQ scores or in-task ratings from scales like the FMS [93]. The SSQ is retrospective and coarse, while continuous ratings can lead the model to classify signals related to the motor action of reporting. A more fundamental issue is the lack of standardized thresholds for defining sickness states. Individual studies frequently devise their own arbitrary cut-offs to convert continuous SSQ scores into binary ("sick" vs. "not sick") or multi-level ("mild," "moderate," "severe") labels. This means that the very definition of the target variable is inconsistent across the literature, severely undermining the comparability of different models and their reported accuracies.

This lack of standardization extends to many aspects of experimental design, creating a profound heterogeneity that hinders scientific synthesis. First, there is significant variability in display technology and the nature of the stimulus. Experiences in fully immersive HMDs are not equivalent to those on large-screen monitors or in CAVE-like projection systems [24], [59], or smartphone based VR systems [37], yet findings are often discussed interchangeably. Similarly, passive viewing of a 360-degree video [79] is fundamentally different from active navigation in an interactive game [19]. Our meta-analysis supports these concerns, revealing statistically significant differences in neural response patterns between different hardware platforms and virtual environment types (see subsubsection IV-F4). Second, experimental protocols are inconsistent, with wide variance in exposure duration, the inclusion and length of rest periods, and the definition of a valid baseline. A particularly problematic practice is the use of an "eyes-closed" baseline, which artificially inflates alpha power and directly confounds the interpretation of alpha suppression, one of the most commonly reported neural markers of cybersickness. Third, the recording itself varies, ranging from research-grade, high-density EEG systems to low-cost, consumer-grade devices with fewer channels and potentially lower signal quality, which may impact the reliability of the derived results [91]. Additionally, reporting quality across studies demonstrates substantial deficiencies in critical methodological parameters. For example, EEG reference electrode configuration is reported in only 49.4% of studies, and notch filter frequency in 22.9%, hampering reproducibility and meta-analytic synthesis. A quantitative analysis of the included literature reveals a reliance on low-density EEG systems, with 82.9% of studies employing 32 or fewer electrodes, and 21.4% employing 10 or fewer electrodes. While such systems are appropriate for applied BCI paradigms that target well-characterized and localized neural signals, their use in the context of cybersickness research presents a significant methodological limitation. The neural underpinnings of cybersickness are not yet fully elucidated, and the research objective for many studies remains fundamentally exploratory. Low-density electrode montages have insufficient spatial resolution for reliable source localization and are blind to activity originating from large, un-sampled cortical areas. This practice, therefore, risks generating an incomplete understanding of cybersickness neurophysiology, potentially overlooking critical contributions from distributed neural networks and biasing the literature towards the few cortical sites that are consistently measured.

Finally, the field suffers from a degree of conceptual imprecision. A significant conceptual issue is the frequent absence of an explicit theoretical framework guiding experimental design and interpretation. Many of the reviewed studies adopt a data-driven, correlational approach, identifying statistical relationships between physiological signals and subjective sickness reports without explicitly stating which theory of cybersickness, such as Sensory Conflict Theory or Postural Instability Theory, they are operating under. This lack of theoretical grounding is problematic for scientific synthesis. Without a stated theoretical context, the interpretation of findings becomes ambiguous, hindering meaningful cross-study comparisons. Furthermore, it weakens the basis for generalizing results, as a "neuromarker" identified in a purely correlational manner lacks a principled foundation for predicting its validity in a new VR application that may stress a different underlying causal mechanism. Moreover, the terms "cybersickness," "motion sickness", "VIMS", "simulator sickness" and more are often used interchangeably, despite evidence that they may represent distinct phenomena. There are fundamental differences, such as the fact that motion sickness occurs in the presence of vestibular motion, while cybersickness occurs with visual motion and is related to visually induced motion also known as vection. Those semantic discrepancies can result in measurable differences. For instance, Rebenitsch et al. [5] found that motion sickness and cybersickness present distinct symptom profiles, while Stanny et al. [100] highlight differences between cybersickness and simulator sickness. Furthermore, there is a tendency to label any statistically significant feature as a "neuromarker" without sufficient validation of its specificity, sensitivity, and reliability across different contexts. Compounding these issues is the prevalent use of small sample sizes and demographically biased populations (e.g., young, male university students), which severely limits the statistical power and external validity of the findings [37], [52].

Taken together, this landscape of methodological and conceptual variability poses a significant threat to replicability and impedes the cumulative progress of the field.

B. Future Directions

The successful identification of cybersickness correlates and the development of high-performing classifiers mark a significant first phase of research. However, to transition

from a collection of isolated, proof-of-concept studies to a mature scientific field capable of producing robust, real-world applications, a concerted effort toward methodological rigor and conceptual clarity is required. Based on the challenges identified in this review, we propose the following roadmap for future research.

First, the field must embrace standardization and open science. The profound heterogeneity in stimuli, hardware, and protocols is the single greatest impediment to comparing results and building cumulative knowledge. We advocate for a community-led effort to develop standardized experimental protocols and benchmark datasets. These datasets should encompass a variety of common VR experiences (e.g., passive roller coasters, active navigation) and include high-quality, multimodal data (e.g., EEG, EOG, ECG, head-motion) alongside well-defined subjective labels. The public release of such resources, as pioneered by a few research groups [43], [77], [82], is essential for validating new methods and ensuring fair, direct comparisons between different classification and prediction models. Accompanying this should be a commitment to open-sourcing analysis code and pre-trained models to enhance transparency and facilitate replication.

Second, research must move beyond correlation to causal mechanisms. While the literature has established strong correlations between specific neural patterns and cybersickness, the causal nature of these relationships remains largely unexplored. Future work should employ interventional techniques to probe these connections. For instance, non-invasive brain stimulation methods like transcranial alternating current stimulation (tACS) could be used to exogenously modulate specific neural oscillations (e.g., alpha or theta power) to determine if this directly influences sickness susceptibility or severity. Such studies, when combined with sophisticated computational neuroscience models based on frameworks like predictive coding, can help disentangle epiphenomenal correlates from the core causal drivers of the cybersickness experience.

Third, for BCI systems to be practical, they must be generalizable and validated in ecologically valid contexts. The performance of many current classifiers drops significantly in cross-subject paradigms. Future research should prioritize the development of advanced machine learning techniques, such as transfer learning and domain adaptation, to create models that are robust to inter-subject and inter-session variability without requiring burdensome user-specific calibration. Furthermore, validation must move beyond the laboratory. Studies conducted in more naturalistic, uncontrolled settings, such as the at-home gaming paradigm explored by Moinnereau et al. [46], are crucial for assessing the real-world viability of these systems.

Fourth, there is a critical need to innovate on the measurement of ground truth. The field's heavy reliance on subjective questionnaires is a persistent limitation. Future work should focus on developing and validating objective, implicit behavioral markers of cybersickness that can supplement or, in some cases, replace subjective reports. These could include metrics derived from user behavior within the VR environment, such as changes in task performance, head motion dynamics (e.g., freezing or reduced exploration), or eye-tracking metrics (e.g., blink rate, saccadic velocity). Integrating these behavioral measures with neurophysiological signals could provide a more holistic and objective picture of the user's state.

Finally, the field should adopt longitudinal study designs to investigate adaptation and recovery. The vast majority of studies are single-session experiments, leaving the neurodynamics of how users adapt to VR over time, or "get their VR legs", largely unknown. Multi-session, longitudinal studies are needed to track the neural changes that accompany this adaptation process. Similarly, while a handful of studies have examined the recovery period immediately following exposure [27], [57], a deeper understanding of the neural mechanisms governing the dissipation of symptoms is required to develop effective post-exposure recovery strategies.

VI. Conclusion

The adverse effects of cybersickness remain a significant obstacle to the widespread adoption and prolonged use of Virtual Reality. Brain-Computer Interfaces have emerged as a promising approach for moving beyond subjective reports toward more objective measurement, prediction, and potential mitigation of this debilitating condition. This paper has provided a systematic and critical review of the literature at the intersection of BCI and cybersickness, synthesizing the current state of knowledge and identifying key challenges and opportunities.

Our synthesis suggests an emerging neurophysiological signature of cybersickness, primarily characterized by increased low-frequency (delta, theta) power and a suppression of alpha-band oscillations in the EEG. Spatially precise modalities like fMRI and fNIRS have localized these dynamics to a network of brain regions including the insula, cingulate cortex, and areas involved in visual-vestibular processing. This foundational knowledge has facilitated the development of computational models, many based on deep learning, that show promise in classifying sickness states, though accuracy claims require careful interpretation given methodological limitations. This research has led to the development of early-stage, real-time, closed-loop adaptive systems that attempt to use neural feedback to modify the VR environment and potentially reduce user discomfort.

However, this review also highlights that the field is hampered by significant methodological heterogeneity, a lack of standardized protocols and ground-truth labels, and a frequent reliance on small, demographically-limited samples. These challenges severely limit the generalizability and replicability of many findings and impede the direct comparison of results across studies. The path toward robust, generalizable BCI applications for cybersickness management requires a concerted shift toward open science practices, including the development of benchmark datasets and standardized evaluation metrics. It also necessitates a move from purely correlational findings to causal investigations and the validation of systems in ecologically relevant, real-world contexts.

In conclusion, the integration of BCI with VR represents more than a novel tool for studying cybersickness; it is a critical enabling technology for the future of immersive computing. The development of truly user-aware systems that can implicitly monitor and dynamically adapt to a user's neurophysiological state in real-time may help address one of the most persistent barriers to the widespread adoption of VR. This line of research is essential for paving the way for experiences that are not only more immersive and effective, but also universally comfortable and accessible.

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